CASE STUDY: 04

REDESIGN OF A CONGESTED URBAN INTERSECTION FOR SAFETY AND EFFICIENCY

RECOURSE

THE ASYMMETRIC MEDIAN FLARE STRATEGY

TEAM CODE N26-035

NIRMAAN'26

THE REALITY OF INDIAN INTERSECTIONS

KEY STATISTICS

01 1.73 Lakh road fatalities in 2023 - MoRTH (highest ever recorded)

02 20.4% of all fatalities are pedestrians = 35,221 deaths

03 31.5% of deaths occur on urban roads - only 8.5% of network

04 60%+ of fatal pedestrian crashes occur at intersections (Mukherjee & Mitra, 2021)

WHY INTERSECTIONS ARE THE EPICENTRE

01 Mixed traffic: cars + 2-wheelers + buses + pedestrians share same space

02 No lane discipline - LT, TH, RT all merge in same 2 lanes

03 Signal timing rarely designed analytically - set arbitrarily

04 Zero pedestrian infrastructure at night

"An intersection is not just an engineering problem. It is a life-safety emergency"

PROBLEM STATEMENT

Problem :

A four-leg signalised urban intersection suffers from severe congestion due to a fixed 110-second cycle operating at only 17% of its Webster's optimum of 630 seconds (Not feasible), unprotected right-turn movements generating pedestrian conflicts at 300 pcu/hr design flow, and a 4.0 m raised median providing no pedestrian refuge despite an 18.0 m crossing width - all within a constrained ROW where no outward expansion is possible.

CAPACITY

Optimize signal timings and lane geometry to reduce vehicular delay.

Objective - 1

SAFETY

Ensure safe pedestrian crossing, specifically addressing night-time visibility.

Objective - 2

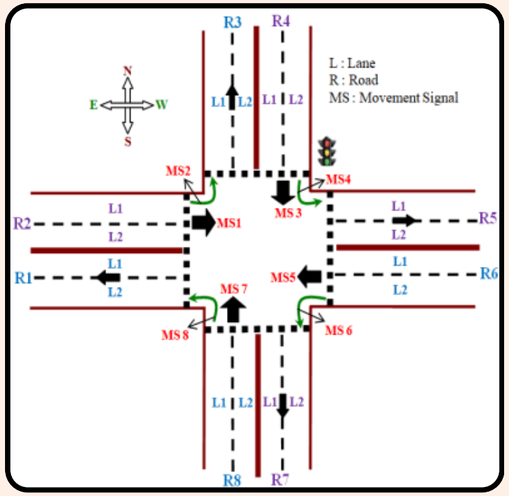
CONSTRAINT

Achieve all improvements within the existing Right-of-Way (ROW) with zero land acquisition.

Objective - 3

EXISTING CONDITION ANALYSIS

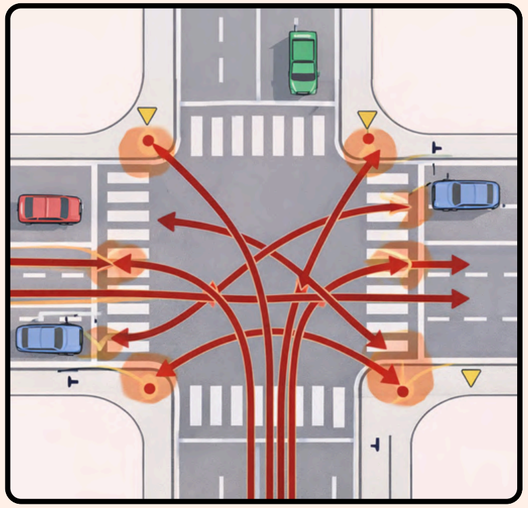
Existing Intersection Overview



Traffic & Pedestrian Analysis



Key Issues Identified



EXISTING INTERSECTION OVERVIEW

The intersection is a four-leg junction with a uniform geometric layout on all approaches.

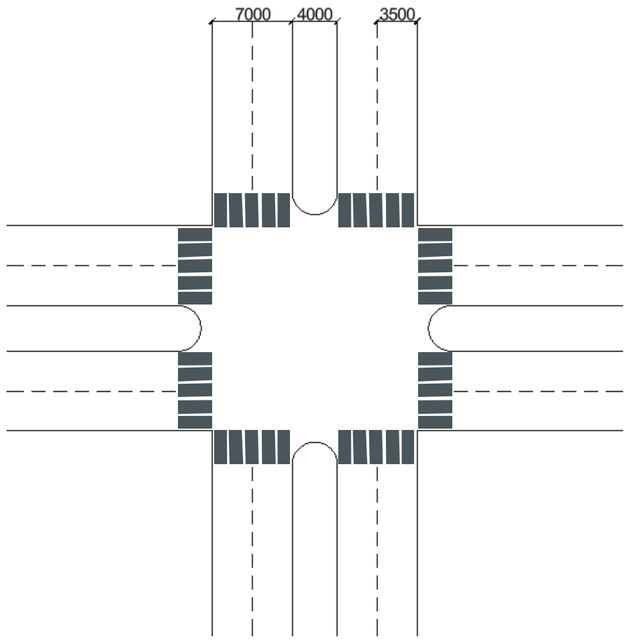
Each approach consists of 2 lanes per direction, with a lane width of 3.5 m, ensuring standard roadway dimensions.

The incoming and outgoing carriageway width is 7.0 m each (2 × 3.5 m), providing balanced flow capacity.

A central median of 4.0 m width is provided along all approaches, aiding in traffic separation and safety.

The total roadway cross-section at the intersection is 18.0 m (7.0 m + 4.0 m + 7.0 m).

Pedestrian crossing width (kerb-to-kerb) is 18.0 m, and the absence of a median refuge island results in a long, uninterrupted crossing distance, reducing pedestrian safety and comfort.



Parameter	Value	Applies To
Lane width	3.5 m	All approaches
Number of lanes per direction	2	All approaches
Incoming carriageway width	2 × 3.5 = 7.0 m	All approaches
Median width	4.0 m	All approaches
Outgoing carriageway width	2 × 3.5 = 7.0 m	All approaches
Total cross-section	7.0 + 4.0 + 7.0 = 18.0 m	All approaches
Pedestrian crossing (kerb-to-kerb)	18.0 m	All arms
Pedestrian refuge island	None	All approaches
Existing cycle length	110 seconds	Fixed-time control
Peak Hour Factor (PHF)	0.8	All movements

TRAFFIC & PEDESTRIAN ANALYSIS

DESIGN FLOW RATES (Q = V / PHF)

Approach	Left Turn	Through	Right Turn	Note
Northbound	170 / 0.80 = 212.5	750 / 0.80 = 937.5	200 / 0.80 = 250.0	Critical P1
Southbound	160 / 0.80 = 200.0	700 / 0.80 = 875.0	240 / 0.80 = 300.0	Critical P3 ★
Eastbound	130 / 0.80 = 162.5	630 / 0.80 = 787.5	200 / 0.80 = 250.0	—
Westbound	150 / 0.80 = 187.5	680 / 0.80 = 850.0	180 / 0.80 = 225.0	Critical P2

Design Flow Formula

$$q = V / PHF$$

where
V = Observed volume
PHF = 0.80 (given)

All values in pcu/hr

Saturation Flow

$$s = 525 \times w$$

pcu/hr/lane

s = 525 × 3.5
= 1,837.5 pcu/hr/lane

Critical Flow Ratios

$$y = q / s$$

y_1 = 937.5 / 1837.5 = 0.510
y_2 = 850.0 / 1837.5 = 0.463
Y = 0.510 + 0.463 = **0.973**
Near breakdown

Signal Design

$$C = \frac{1.5L+5}{1-Y}$$

L = 2 × 4 = 8 s
C = (1.5×8 + 5) / (1 - 0.973) = **630 s**
(Webster breakdown condition)
Not feasible
Given existing C = 110 s

Green Time Split

$$g_i = y_i / Y \times Geff$$

$$Geff = C - L$$

Geff = 110 - 8 = 102 s
g1 = (0.510 / 0.973) × 102 ≈ 53s
g2 = (0.463 / 0.973) × 102 ≈ 49s
Total = 53 + 49 = 102 s

Volume-to-Capacity Ratio

$$v/c = \frac{Y}{1 - (L/C)}$$

v/c = 0.973 / (1 - 8/110)
v/c = 0.973 / 0.927 = **1.050**
→ **OVER CAPACITY**

TRAFFIC & PEDESTRIAN ANALYSIS

Element	Distance	Code
Incoming carriageway (2 × 3.5 m)	7.0 m	Geometry
Median (no refuge island)	4.0 m	Geometry
Outgoing carriageway (2 × 3.5 m)	7.0 m	Geometry
Kerb ramp near side (150 mm kerb × 1:15)	2.25 m	IRC 103:2012
Kerb ramp far side (150 mm kerb × 1:15)	2.25 m	IRC 103:2012
Tactile warning tile — near side	0.60 m	IRC 103
Tactile warning tile — far side	0.60 m	IRC 103
TOTAL EFFECTIVE CROSSING DISTANCE	23.7 m	

Normal Pedestrian (1.2 m/s)

Crossing time
= 23.7 / 1.2 = 19.75 s
Start-up delay
= 3.00 s
Clearance buffer
= 3.00 s
Min green needed = 25.75 ≈ 26 s

Elderly Pedestrian (0.8 m/s)

Crossing time
= 23.7 / 0.8 = 29.63 s
Start-up delay
= 3.00 s
Clearance buffer
= 3.00 s
Min green needed = 35.63 ≈ 36 s

Available Green vs Required

C = 110 s | L = 8 s | Geff = 102 s
g_1 (N–S) = 53 s | g_2 (E–W) = 49 s

	Normal	Elderly
Min green needed :	26 s	36 s
g_1 available (N–S) :	53 s ✓	53 s ✓
g_2 available (E–W) :	49 s ✓	49 s ✓

Safety Verdict

- **Timing** : Adequate for both normal and elderly
- **Single-stage** : 23.7 m uninterrupted **- no refuge**
- **Turning conflict** : No phase separation from turning vehicles
- **Night safety** : No raised refuge, no bollards, no ped heads
- **IRC SP:41** : **Refuge island MISSING - non-compliant**

Timing passes. Geometry and safety FAIL.

KEY ISSUES IDENTIFIED

- Unprotected Right
- Cycle Length Crisis
- No Pedestrian Refuge

Issues	IRC Code	Requirement	Existing Condition (Violation)
V-1	IRC SP:41 8.3	Protected RT required if RT > 200 pcu/hr	SB-RT = 300 pcu/hr Unprotected → pedestrian conflict
V-2	Webster's Formula	Cycle length ≈ optimum	Existing condition: Webster's optimal C = 630 s Given existing C = 110 s
V-3	IRC SP:44 / IRC 103:2012	Pedestrian must cross within green time	Direct pedestrian conflict → Timing passes, geometry and safety FAIL

Intersection grossly under-cycled Y = 0.973, v/c = 1.050 → over capacity Causes extreme vehicular delay and queue buildup

OUR SOLUTION

01. **Geometric Design : Asymmetric Median Flare Geometry**

02. **Signal Design & Traffic Control - 3 phase signal**

03. **Design Innovations : Pedestrian & Night-Time Safety**



GEOMETRIC DESIGN - ASYMMETRIC MEDIAN FLARE GEOMETRY

01

The New Carriageway Cross-Section

- 3×3.0 m incoming lanes
- 2×3.5 m outgoing
- 2.0 m median as pedestrian refuge



Asymmetric Median Flaring

- Median reduced 4.0 m → 2.0 m
- 2.0 m allocated to incoming side



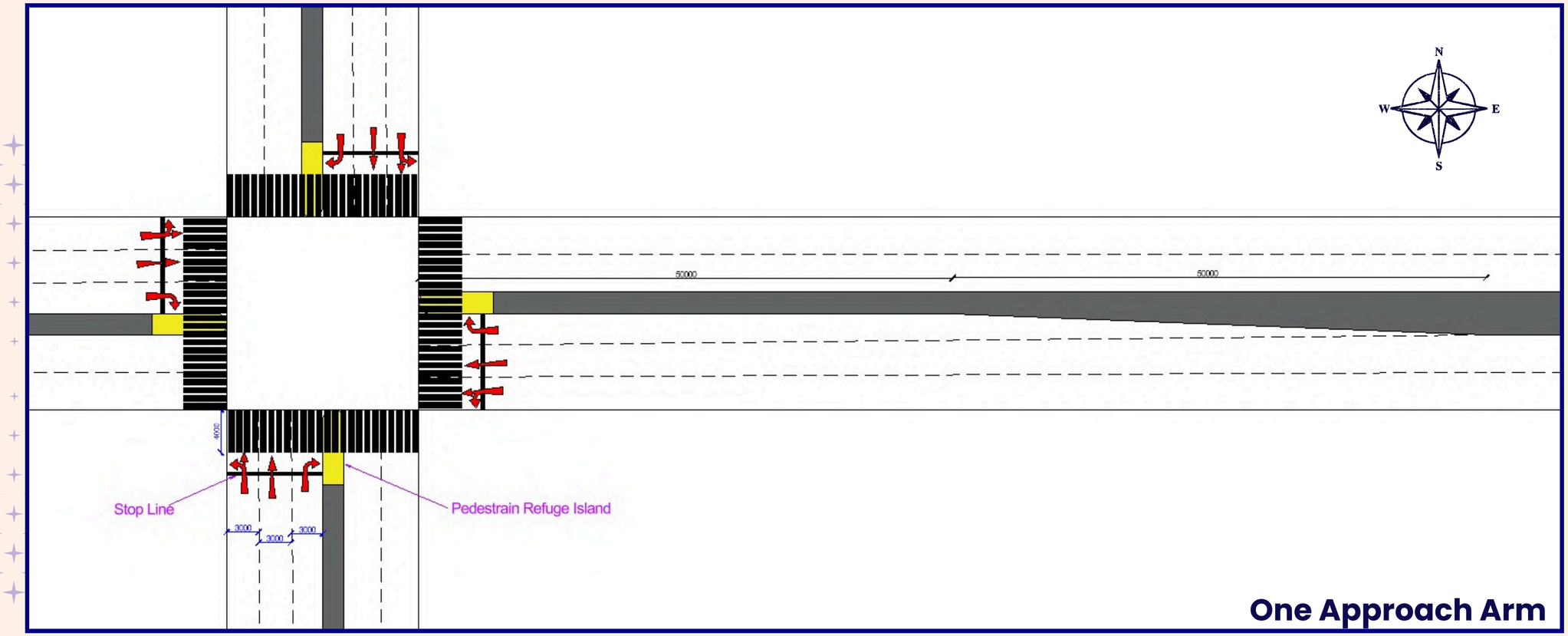
The Core Constraint: Zero Land Acquisition

- Design confined within 18.0 m ROW
- No widening, no demolition



PROPOSED INTERSECTION DESIGN

Asymmetric median flaring with optimized signal and safety interventions



One Approach Arm

IRC SP:41:1994

$L_{des} = W \times V / 2.0$
$= 2.0\ m \times 50\ km/h / 2.0$
$= 50.0\ m$ ✓

$L_{min} = W \times V / 3.6$
$= 2.0 \times 50 / 3.6$
$= 27.8\ m$ (minimum required)

Provided 50 m > 27.8 m ✓
EXCEEDS MINIMUM
Taper ratio = 1 : 25

NEW CARRIAGEWAY CROSS-SECTION

Cross-Section Allocation

- Incoming (9.0 m): 3 lanes × 3.0 m
- Median (2.0 m): Pedestrian refuge island
- Outgoing (7.0 m): 2 lanes × 3.5 m

Total Width

9.0 + 2.0 + 7.0 = 18.0 m

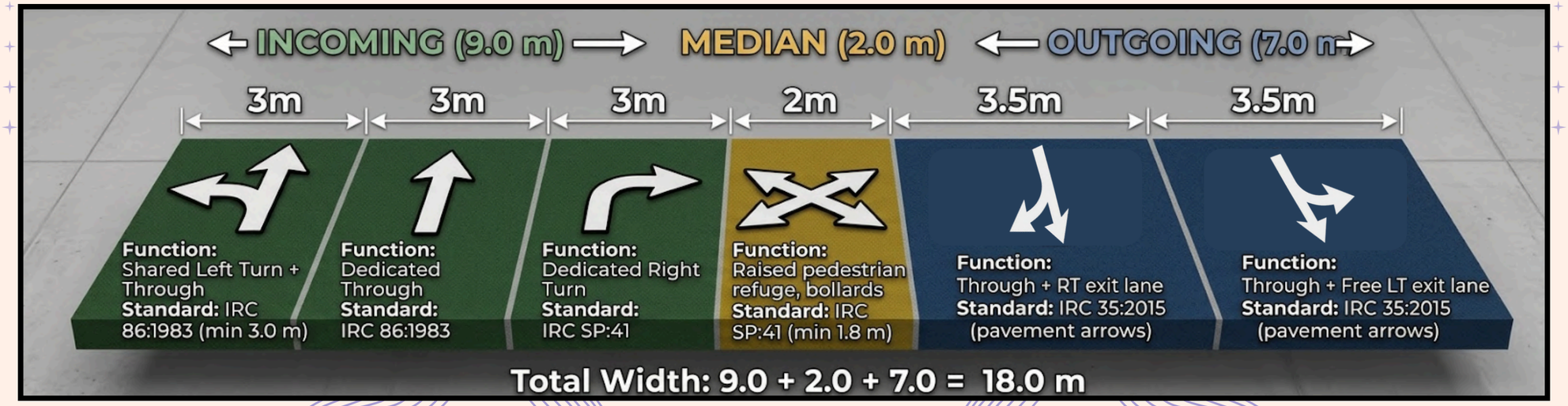
No additional Right-of-Way required



ASYMMETRIC MEDIAN FLARE GEOMETRY

Lane Configuration

Lane	Width	Function	IRC Code
Lane 1 (outer)	3.0 m	Shared Left Turn + Through	IRC 86:1983 (min 3.0 m) ✓
Lane 2 (middle)	3.0 m	Dedicated Through	IRC 86:1983 ✓
Lane 3 (inner)	3.0 m	Dedicated Right Turn	IRC SP:41 ✓
Median / Refuge Island	2.0 m	Raised pedestrian refuge, bollards	IRC SP:41 (min 1.8 m) ✓
Outgoing Lane 1	3.5 m	Through + Free LT exit lane	IRC 35:2015 (pavement arrows)
Outgoing Lane 2	3.5 m	Through + RT exit lane	IRC 35:2015 (pavement arrows)



SIGNAL DESIGN & TRAFFIC CONTROL

02

Signal Design – Webster's Optimum Method

- Cycle length optimized using flow ratios (Y)
- Reduced from 110 s → 96s (-13% delay)



Signal Phase & Traffic Control

- 3-phase operation (N-S, E-W, Protected RT)
- Balanced green time + safe pedestrian crossing



SIGNAL DESIGN - WEBSTER'S OPTIMUM METHOD

DESIGN FLOW RATES (Q = V / PHF)

Peak Hour Factor (PHF) = 0.80 applied to all observed volumes to obtain design flow rates

Approach	Left Turn	Through	Right Turn	Note
Northbound	170 / 0.80 = 212.5	750 / 0.80 = 937.5	200 / 0.80 = 250.0	Critical P1
Southbound	160 / 0.80 = 200.0	700 / 0.80 = 875.0	240 / 0.80 = 300.0	Critical P3 ★
Eastbound	130 / 0.80 = 162.5	630 / 0.80 = 787.5	200 / 0.80 = 250.0	-
Westbound	150 / 0.80 = 187.5	680 / 0.80 = 850.0	180 / 0.80 = 225.0	Critical P2

Design Flow Formula

q = V / PHF

where

V = Observed volume

PHF = 0.80 (given)

All values in pcu/hr

★ SB-RT design flow = 300 pcu/hr > IRC SP:41 threshold of 200 pcu/hr. Protected right-turn is legally mandatory.

SIGNAL DESIGN - WEBSTER'S OPTIMUM METHOD

Saturation Flow

Lane: 3.5m → 3.0m

s = 525 × 3.0 = 1,575 pcu/hr/lane (incoming lane width = 3.0 m)

01

Critical Flow Ratios

Phase 1 (E–W through, 2 lanes):
q_1 = (680/0.80) / 2 = 425.0
y_1 = 425.0 / 1575 = 0.270
Phase 2 (N–S through, 2 lanes):
q_2 = (750/0.80) / 2 = 468.8
y_2 = 468.8 / 1575 = 0.298
Phase 3 (All turns protected, 1 lane):
q_3 = (240/0.80) / 1 = 300.0
y_3 = 300.0 / 1575 = 0.190
Y = 0.270+0.298+0.190 = 0.758

Y < 1.0 → Intersection UNSATURATED

02

Webster's Optimum Cycle Length

Start-up: 2.0 s (IRC SP:41)
Clearance: 2.0 s (compact geometry)
L = 3 phases × 4.0s = 12.0 s
Total Lost Time = 12 s

C = (1.5×12 + 5) / (1 – 0.758)
C = 23 / 0.242 = **95 s**

03

Green Time Allocation

Geff = 95 - 12 = 83 s

- g_1 = (0.270/0.758) × 83 = 29.6 ≈ 30 s (E–W walk)
- g_2 = (0.298/0.758) × 83 = 32.6 ≈ 33 s (N–S walk)
- g_3 = (0.190/0.758) × 83 = 20.8 ≈ 21 s (All turns)

Total = 84 s

04

Capacity Check (v/c ratio)

v/c = Y ÷ (1 - L/C)
v/c = 0.758 / (1 - 12/95)
v/c = 0.758 / 0.874
v/c = 0.867
v/c < 1.0 → Within Capacity
Excellent LOS ROW-constrained urban intersection

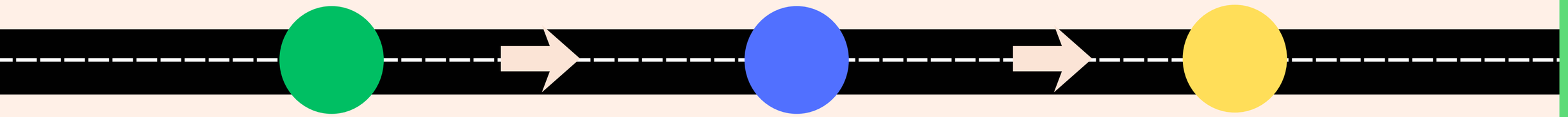
05

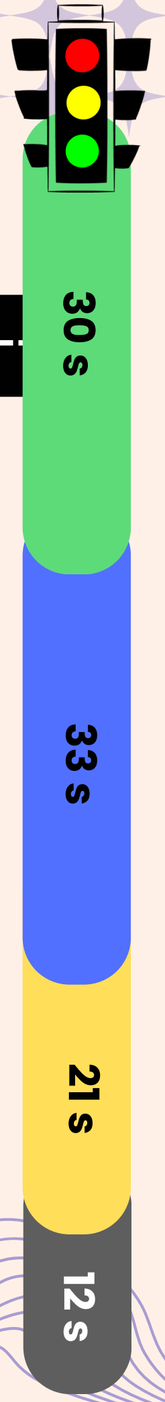
Adopted cycle = 96s

Y = 0.758 (unsaturated) v/c = 0.867 13% reduction from existing 110s

Intersection operating well within capacity

SIGNAL PHASE & TRAFFIC CONTROL





Phase 1 (Left Block)

E-W Through

30 s

E–W WALK

Phase 2 (Middle Block)

N-S Through

33 s

N–S WALK

Phase 3 (Right Block)

Right Turns

21 s

All Pedestrians RED

Protected RT

Movement	Phase 1	Phase 2	Phase 3
EB / WB through	GO	RED	RED
NB / SB through	RED	GO	RED
All LT (NB/SB/EB/WB)	FREE LEFT	FREE LEFT	FREE LEFT
All RT (NB/SB/EB/WB)	RED	RED	GO (protected)
E-W pedestrians	WALK	RED	RED
N-S pedestrians	RED	WALK	RED

Free Left Turn

- Allowed in all phases
- Blinking amber signal (proceed with caution)
- Yield to traffic & pedestrians
- Improves flow, reduces delay

DESIGN INNOVATIONS

03

Pedestrian Safety

- Refuge Island
- Protected Right Turn
- Blinking Amber LT Arrow
- Staggered Stop Lines for Heavy Vehicles
- Yellow Box Junction
- Raised Table-Top Crossings

Night-Time Safety

- Four-Layer Night-Time Safety System
  1. Photoluminescent Concrete
  2. Solar LED Road Studs
  3. Thermoplastic Markings
  4. High-Mast LED Lighting

PEDESTRIAN SAFETY

Refuge Island

- Provides a safe waiting space for pedestrians while crossing
- Divides crossing into two shorter stages, reducing risk
- Improves safety for elderly and slow pedestrians
- Minimizes conflicts with turning vehicles
- A 2.0 m refuge island is proposed to enhance crossing safety



Scenario	Time Needed	Green Available	Result
Average pedestrian, N-S, single stage (1.2 m/s)	26 s	33 s	PASS (+7 s)
Average pedestrian, E-W, single stage (1.2 m/s)	26 s	30 s	PASS (+4 s)
Elderly, N-S, single stage (0.8 m/s)	36 s	33 s	FAIL — use island
Elderly, E-W, single stage (0.8 m/s)	36 s	30 s	FAIL — use island
Elderly, N-S, Stage 1 via island (0.8 m/s)	20 s	33 s	PASS (+13 s)
Elderly, E-W, Stage 1 via island (0.8 m/s)	20 s	30 s	PASS (+10 s)
Elderly, N-S, Stage 2 via island (0.8 m/s)	20 s	33 s	PASS (+13 s)
Elderly, E-W, Stage 2 via island (0.8 m/s)	20 s	30 s	PASS (+10 s)

Average pedestrians pass in single-stage crossing, but elderly pedestrians fail due to insufficient green time
With refuge island (two-stage crossing), all cases pass with adequate safety margins

PEDESTRIAN SAFETY



Protected Right Turn

- All RT vehicles move only during Phase 3
- Eliminates unprotected turning conflicts and reduces crash risk

Setback Stop Line for Heavy Vehicles

- Stop line set back by 3.0 m to create internal buffer
- Prevents heavy vehicle encroachment into crossing
- Accommodates bus/truck swept path safely

Blinking Amber LT Arrow

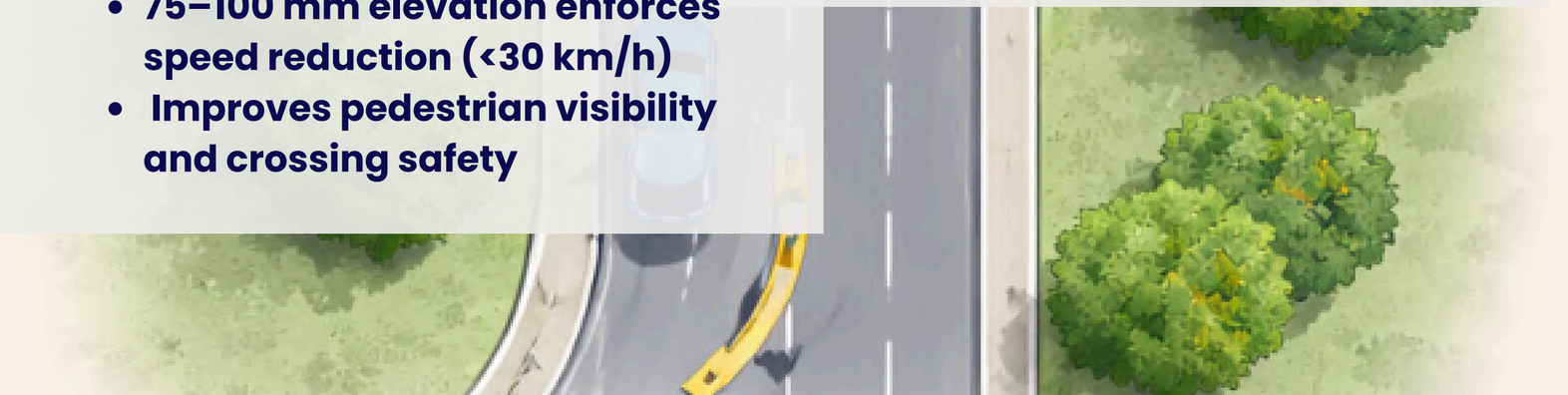
- LT vehicles operate on continuous yield control, strictly prioritizing pedestrian right-of-way.
- Eliminates the need for physical slip lanes while mitigating conflicts through visually enforced caution

Yellow Box Junction (ICAM)

- Prevents vehicles from entering blocked intersection
- Eliminates gridlock and maintains flow continuity

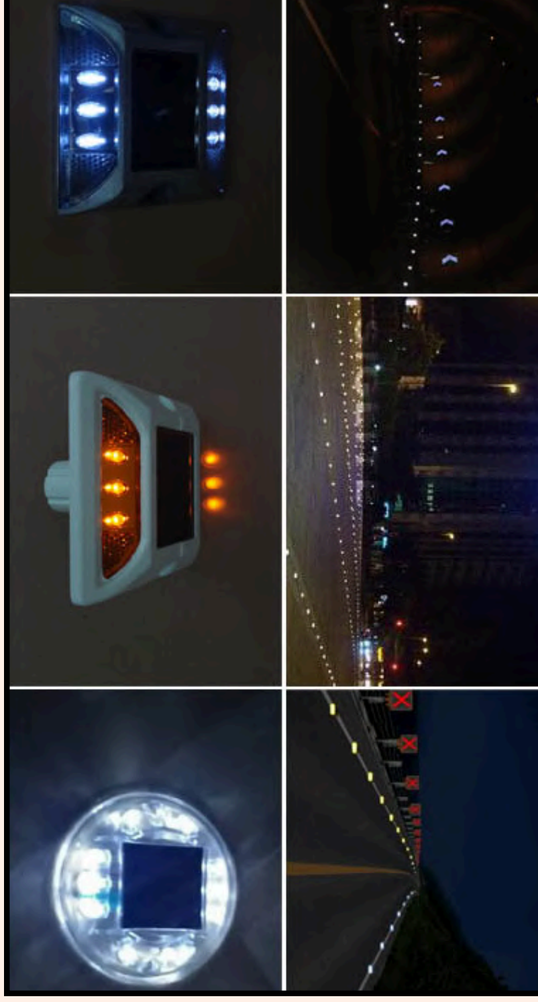
Raised Table-Top Crossings

- 75–100 mm elevation enforces speed reduction (<30 km/h)
- Improves pedestrian visibility and crossing safety





Thermoplastic Markings



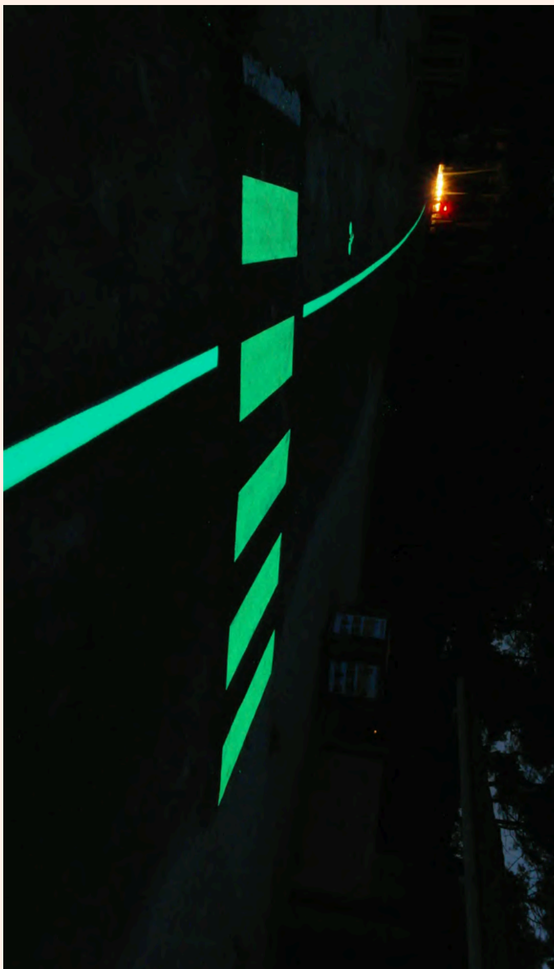
Solar LED Road Studs

NIGHT-TIME SAFETY

High-Mast LED Lighting



Photoluminescent Concrete



REAL-WORLD VALIDATION

Our Design Is Already Proven in India

LBS MARG, MUMBAI

Bloomberg BIGRS · 2019–2022

BEFORE: 13 pedestrian deaths per year on 10 km urban corridor

INTERVENTION: Raised median refuge islands · Lane narrowing 3.0–3.5 m · Retroreflective markings

RESULT: **Fatalities → NEAR ZERO**

✓ Validates: 3.0 m lanes · Refuge island · Thermoplastic markings

PUNE SMART CITY

Aundh & ITI Road · Smart City Mission

BEFORE: Pedestrians hit by fast-turning vehicles at intersections

INTERVENTION: Raised table-top intersections — 75–100 mm elevation · No land acquisition

RESULT: **Near-miss conflicts reduced significantly**

✓ Validates: Raised table-top crossing specification

DELHI ASHRAM CHOWK

Delhi Traffic Police · Major Arterials

BEFORE: Heavy DTC buses could not turn without encroaching on corner property

INTERVENTION: Stop lines shifted back to create 3.0 m swept path buffer zone

RESULT: **Bus conflicts eliminated — zero demolition required**

✓ Validates: Staggered stop lines for WB-12 heavy vehicles

BENGALURU IT CORRIDOR

Smart City Pilot · 2022

BEFORE: Zero night-time visibility on pedestrian paths during power outages

INTERVENTION: Photoluminescent $SrAl_2O_4$:Eu,Dy concrete on pedestrian paths

RESULT: **Survived 2 monsoon seasons — zero material degradation**

✓ Validates: Photoluminescent concrete durability in Indian conditions

Every design element has a documented Indian implementation with measured results **4 CASE STUDIES · 4 VALIDATIONS**

VALIDATION & COMPARISION

Full IRC Compliance Check



Final Performance Comparison



IRC COMPLIANCE AUDIT

IRC Code	Requirement	Provided	Status
IRC 86:1983	Min lane width 3.0 m	**3.0 m**	PASS ✓
IRC SP:41	Min taper 27.8 m	**50.0 m**	PASS ✓
IRC SP:41 §8.3	Refuge island min 1.8 m	**2.0 m**	PASS ✓
IRC SP:41	Protected RT > 200 pcu/hr	**Phase 3 @ 21 s**	PASS ✓
IRC:93 §8.1	Blinking Amber permissive LT	**Amber LT Arrow provided**	PASS ✓
IRC:93	All 5 signal warrants	**All 5 satisfied**	PASS ✓
IRC 35:2015	Road markings + signs	**Grade A glass beads**	PASS ✓
IRC 35 §9	Yellow box junction	**100 mm hatching**	PASS ✓
IRC 103:2012	Kerb ramps 1:15 slope	**Both ends provided**	PASS ✓
IRC 103:2012	Refuge width min 2.0 m	**2.0 m with bollards**	PASS ✓
IRC SP:72	50 lux at crossings	**150 W LED; 50 lux**	PASS ✓

FINAL PERFORMANCE COMPARISON

Parameter	Existing Design	Proposed Design	Change
Signal cycle length	110 s	96 s	-13%
Webster’s optimum	630 s (110 s used)	95 s	Rational
Signal phases	2 (no protected turns)	3 (protected RT)	1
Total flow ratio Y	0.973 (near saturated)	0.758	Unsaturated
v/c ratio	1.050 (oversaturated)	0.867	Within capacity
Saturation flow/lane	1,837.5 pcu/hr (3.5 m)	1,575 pcu/hr (3.0 m)	-14% / lane
Phase 1 green (E–W)	49 s	30 s	Structured
Phase 2 green (N–S)	53 s	33 s	Structured
Phase 3 green (turns)	None	21 s	Added
Incoming lanes	2 × 3.5 m = 7.0 m	3 × 3.0 m = 9.0 m	+2.0 m

Parameter	Existing Design	Proposed Design	Change
Median / refuge	4.0 m (no refuge)	2.0 m (raised island)	Island added
Total cross-section	18.0 m	18.0 m	Zero ROW
Avg ped N–S (26 s)	26 s < 53 s - PASS ✓	26 s < 33 s - PASS ✓	Structured
Avg ped E–W (26 s)	26 s < 49 s - PASS ✓	26 s < 30 s - PASS ✓	Structured
Elderly single-stage	36 s < 53 s - PASS ✓	36 s > 33 s - FAIL ✓	Island needed
Elderly two-stage	IMPOSSIBLE (no island)	20 s < 33 s - PASS ✓	Enabled
RT–pedestrian conflict	300 pcu/hr unprotected	Fully eliminated	ELIMINATED
LT–pedestrian conflict	Unprotected all phases	Free flowing with Blinking Amber	ELIMINATED
Night safety	None	4-layer enhanced lighting	Added
Land acquisition	—	ZERO	Zero cost

DESIGN ITERATIONS: WHY CHOOSE THIS DESIGN

ELIMINATING ALTERNATIVES THROUGH MATHEMATICAL AND SPATIAL STRESS-TESTING.

✗ CONVENTIONAL FREE-LEFT SLIP LANE

The Fatal Flaw: Requires physical corner channelisation islands.
The Result: Demands building/footpath demolition, strictly violating the Zero-ROW constraint.

✗ THE 45-SECOND RCUT (U-TURN) SYSTEM

The Fatal Flaw: Fails the heavy vehicle swept-path test.
The Result: A WB-12 bus needs a 25.0 m diameter to U-turn. Our road only provides 11.0 m, which would force a 3-point reverse turn and cause total arterial gridlock

✗ NARROWING LEFT LANE TO 2.0M

The Fatal Flaw: Violates the IRC absolute minimum (2.75m) for cars.
The Result: Leaves only one lane for the massive 937.5 pcu/hr Through traffic, instantly forcing the intersection into oversaturation (Y > 1.0).

✗ SYMMETRICAL MEDIAN SQUEEZE

The Fatal Flaw: Giving 1.0m of reclaimed space to the outgoing carriageway.
The Result: Wastes valuable urban space, as outgoing vehicles do not queue.

✓ **The Asymmetric Median Flare is the only geometric solution that mathematically satisfies capacity demands while strictly adhering to the Zero-ROW mandate.**

COST–BENEFIT ANALYSIS

Maximum Safety. Minimum Expenditure. Zero Land Cost.

WHAT THIS DESIGN COSTS

Item	Cost Level
Road restriping — 4 arms × 50 m taper + lane arrows	Very Low
Median reconstruction 4.0 m → 2.0 m	Moderate
Refuge island — 4 crossings	Moderate
Raised table-top crossings — 8 crossings	Moderate
Signal heads upgrade — LED + amber LT arrow	Moderate
Photoluminescent concrete — zebra strips only	+15% only
Solar road studs — 16 units	Low
High-mast LED poles — 4 units	Moderate
Yellow box junction marking	Negligible
Staggered stop line repainting	Negligible
Land acquisition	ZERO
Buiiding demolition	ZERO

WHAT THIS DESIGN DELIVERS

Benefit	Value
Signal cycle length	110 s → 96 s (-13%)
Total flow ratio Y	0.95 → 0.7579 ✓
v/c ratio	>1.0 → 0.848 ✓
RT–pedestrian conflict	FULLY ELIMINATED
Elderly pedestrian crossing	Safe 2-stage via island
Night visibility	3 grid-independent layers
Land acquisition cost	₹ 0
Building demolition	₹ 0
IRC SP:88 Safety Audit	13 PASS
Construction timeline	4–6 weeks only

How RECOURSE Compares to Alternatives

Solution Type	Approx. Cost	ROW Needed	Time to Build	Safety
Grade Separator / Flyover	₹ 50–150 Crore	Major land acquisition	3–5 years	High
Full Road Widening	₹ 10–30 Crore	Property demolition	1–2 years	High
Signal Re-Timing Only	₹ 5–10 Lakh	None	1 week	Low — no geometry
★ **RECOURSE (This Design)**	**₹ 50–80 Lakh**	**ZERO**	**4–6 weeks**	**HIGH ✓**

RECOURSE delivers flyover-level safety outcomes at **1% of flyover cost** — built in 4 weeks, within the existing 18.0 m road.

COST
₹ 50–80 Lakh
Zero land cost

SPEED
4–6 weeks to build
No ROW approvals

SAFETY
IRC SP:88 — 13/14 PASS
Flyover-equivalent outcome

CONCLUSION

PRECISION ENGINEERING WITHIN EXISTING BOUNDARIES

THE 5 PROBLEMS WE WERE GIVEN		THE 5 SOLUTIONS DELIVERED
Severe traffic congestion and long vehicular delay **LOS F**	01	Cycle length: 110 s → 96 s (13% reduction) Webster's Y = 0.758. v/c = 0.867 2-phase → 3-phase. **LOS D**
Inadequate pedestrian safety (no refuge island)	02	Pedestrian: 2.0 m refuge island. Every scenario passes. Elderly fully protected via 2-stage crossing
Unprotected RT movements	03	Phase 3: All 4 RT movements protected. RT pedestrian conflict fully eliminated
Constrained ROW zero land available	04	ROW: 18.0 m in, 18.0 m out. Zero land acquisition. Zero building demolition
Night-time pedestrian safety a critical concern	05	Night safety: 4 layers. 3 grid-independent. Photoluminescent concrete. Solar studs. LED

REFERENCES

A. Indian Roads Congress (IRC) Codes and Special Publications

- IRC SP:41–1994

Guidelines for the Design of At-Grade Intersections in Rural & Urban Areas, Indian Roads Congress.

- IRC SP:44–2017

Guidelines for Traffic Signals on Urban Roads, Indian Roads Congress.

- IRC 103–2012

Guidelines for Pedestrian Facilities, Indian Roads Congress.

- IRC SP:117–2018

Guidelines for Pedestrian Safety Facilities, Indian Roads Congress.

- IRC 86–1983

Geometric Design Standards for Urban Roads in Plains, Indian Roads Congress.

- IRC 35–2015

Code of Practice for Road Markings, Indian Roads Congress.

- IRC 106–1990

Guidelines for Capacity of Urban Roads in Plain Areas, Indian Roads Congress.

B. Signal Design - Foundational References

Webster, F.V. (1958)

- Traffic Signal Settings, Road Research Technical Paper No. 39, UK.


WE ARE OPEN FOR THE QUESTIONS NOW!!
Thank You!

APPENDIX: IRC CODE REFERENCES & PROVISIONS USED

1. IRC 86:1983 — Geometric Design Standards for Urban Roads

Relevant Clause: Lane Width

Requirement: Minimum lane width = 3.0 m

Purpose:

Ensures safe maneuverability of vehicles

Reduces side friction and conflict

Application in your design:

Adopted 3.0 m lane width for all approaches → meets minimum standard

2. IRC SP:41 — Guidelines for Design of At-Grade Intersections

(a) Taper Length

Requirement: Minimum taper length depends on speed & width (≈ 1:15 to 1:20)

Your case: 27.8 m minimum

Purpose:

Smooth merging/diverging movement

Prevents abrupt lane shifts

(b) Protected Right Turn (RT)

Condition: When RT volume > 200 PCU/hr

Provision:

Separate signal phase required

Your design:

Phase 3 provided (21 sec) → Correctly justified

3. IRC SP:41 §8.3 — Refuge Islands

Requirement: Minimum width = 1.8 m

Purpose:

Pedestrian safety during crossing

Two-stage crossing provision

Your design:

Provided 2.0 m refuge island → Safer than minimum

4. IRC 93 — Guidelines on Traffic Signals

(a) Permissive Left Turn (LT)

Provision:

Allowed with blinking amber arrow

Purpose:

Improves efficiency without full signal phase

Your design:

Implemented Amber LT Arrow

(b) Signal Warrants

5 Warrants include:

Traffic volume

Pedestrian volume

Accident history

Delay

Combination conditions

Your design:

All 5 satisfied → Justifies signalization

5. IRC 35:2015 — Road Markings

(a) Marking Material

Requirement: High visibility thermoplastic/paint

Your design:

Grade A glass beads used → High retroreflectivity

(b) Yellow Box Junction (§9)

Purpose:

Prevent vehicles blocking intersection

Requirement:

Diagonal yellow grid marking

Your design:

100 mm thick hatching provided

6. IRC 103:2012 — Pedestrian Facilities

(a) Kerb Ramps

Requirement:

Slope = 1:12 (preferred) to 1:15 acceptable

Purpose:

Accessibility (wheelchair, elderly)

Your design:

Provided at both ends of crossing

(b) Refuge Width

Requirement: Minimum 2.0 m

Your design:

2.0 m with bollards → Added safety

7. IRC SP:72 — Street Lighting

Requirement:

Minimum illumination at crossings = 50 lux

Purpose:

Night visibility for pedestrians and drivers

Your design:

150 W LED achieving 50 lux → compliant